FACULTY OF COMPUTING

Semester 01 Session 2024/2025

Subject	**:**	**Database Engineering (SCST1143)**
Task	**:**	**Phase 2 (P2) – Database Conceptual Design (ERD)**
Due	**:**	**WEEK 07 (17 – 23 November 2024) – draft (as a weekly task) (2.5%)** **WEEK 09 (01 – 07 December 2024) (5%)**

P2 Guidelines

No	Task	Deliverables (Items in Database)
1.	• Based on the requirements in P1, identify the entities and attributes of each entity. • Determine all possible relationships between entities. • Determine the multiplicities for each relationship. • Create a conceptual ERD to represent the information above. • Include enhanced ERD features wherever possible. • Produce the data dictionary for the created conceptual design.	DB conceptual design: a) Proposed business rule b) Proposed data & transactional c) Conceptual ERD (minimum 5 tables) d) Enhanced ERD (EERD) e) Data dictionary

Database Conceptual Design

Prepare a Database conceptual design as below:

1.0 Introduction
2.0 Flowchart (Proposed System)
3.0 Data & Transaction requirement
  3.1 Proposed business rule
  3.2 Proposed data & transactional
4.0 Database conceptual design
  4.1 Conceptual ERD
  4.2 Enhanced ERD (EERD)
5.0 Data dictionary
6.0 Summary

P2 Rubrics

1. **Reporting (2%)**

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Document the whole process of designing the ERD.	Documented **ALL** reports on the construction of the database design. Shows **FULL** understanding of the requirements based on the case study.	Documented **MAJOR** reports on the construction of the database design. Shows a **GOOD** understanding of the requirements based on the case study.	Documented **MINIMAL** reports on the construction of the database design. Shows **MINIMAL** understanding of the requirements based on the case study.	**LITTLE** evidence that the documents are based on the given case study.

2. **ERD Design Process (3%)**

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Produces the necessary primary keys (PK) and/or foreign keys (FK)	Correctly choose **ALL** the PKs and FKs, about the naming convention	Correctly choose **MOST** of the PKs and FKs, about the naming convention.	Correctly choose **SOME** of the PKs and FKs, about the naming convention.	**WRONGLY** choose the PKs and FKs.
Relates the entities and determines the multiplicity of each relationship using UML notation	Correctly interpreted **ALL** business rules with **a CLEAR** idea of the entities and multiplicity of relationships	Correctly interpreted **MOST** of the business rules with a **GOOD** idea of the entities and multiplicity of	**INACCURATELY** interpreted the business rules with the **WRONG** idea on the entities and multiplicity of relationships involved in	**UNABLE** to interpret the business rules with **NO IDEA** on the entities and multiplicity of relationships involved in the design using

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
	involved in the design using UML notations.	relationships involved in the design using UML notations.	the design using UML notations.	UML notations.
Illustrates the database design in a complete ERD using UML notation	A **COMPLETE** populated the tables with **CORRECT** data elements as outlined in the ERD design.	Populated the tables with **MAJORITY** data elements as outlined in the ERD design - with MINIMAL errors.	Populated the tables with **MINIMAL** data elements as outlined in the ERD design - with **MINIMAL** errors.	Populated the tables with **MINIMAL** data elements as outlined in the ERD design - with **NUMEROUS** errors.
Identifies all the basic concepts in ERD (entity, attribute, relationship)	**COMPLETE** understanding of the basic concepts in **ERD**	**ALMOST** complete understanding of the basic concepts in **ERD.**	Exhibit a **MINIMAL** understanding of the basic concepts in **ERD**.	**UNABLE** to show understanding of the basic concepts of **ERD.**
Represents the main entities and attributes based on the given case study	**Completely** and **accurately** use of names, definitions in entities and attributes – **ALL** table names are appropriately used regarding its data elements	Some **MINOR** inaccurate use of names, definitions in entities, and attributes - **MOST** table names are not clear regarding their data elements	**INACCURATE** the use of names, definitions in entities and attributes - table name **DID NOT** correlate with its data elements	**WRONG** the use of names, and definitions in entities and attributes. Shows **NO** understanding in identifying the tables based on the case study.